ENGR489 PROJECT · VICTORIA UNIVERSITY OF WELLINGTON

HUNTING MR. X
WELLINGTON EDITION

A new Open Day demo:

a classic board game, played online on a real map of Wellington

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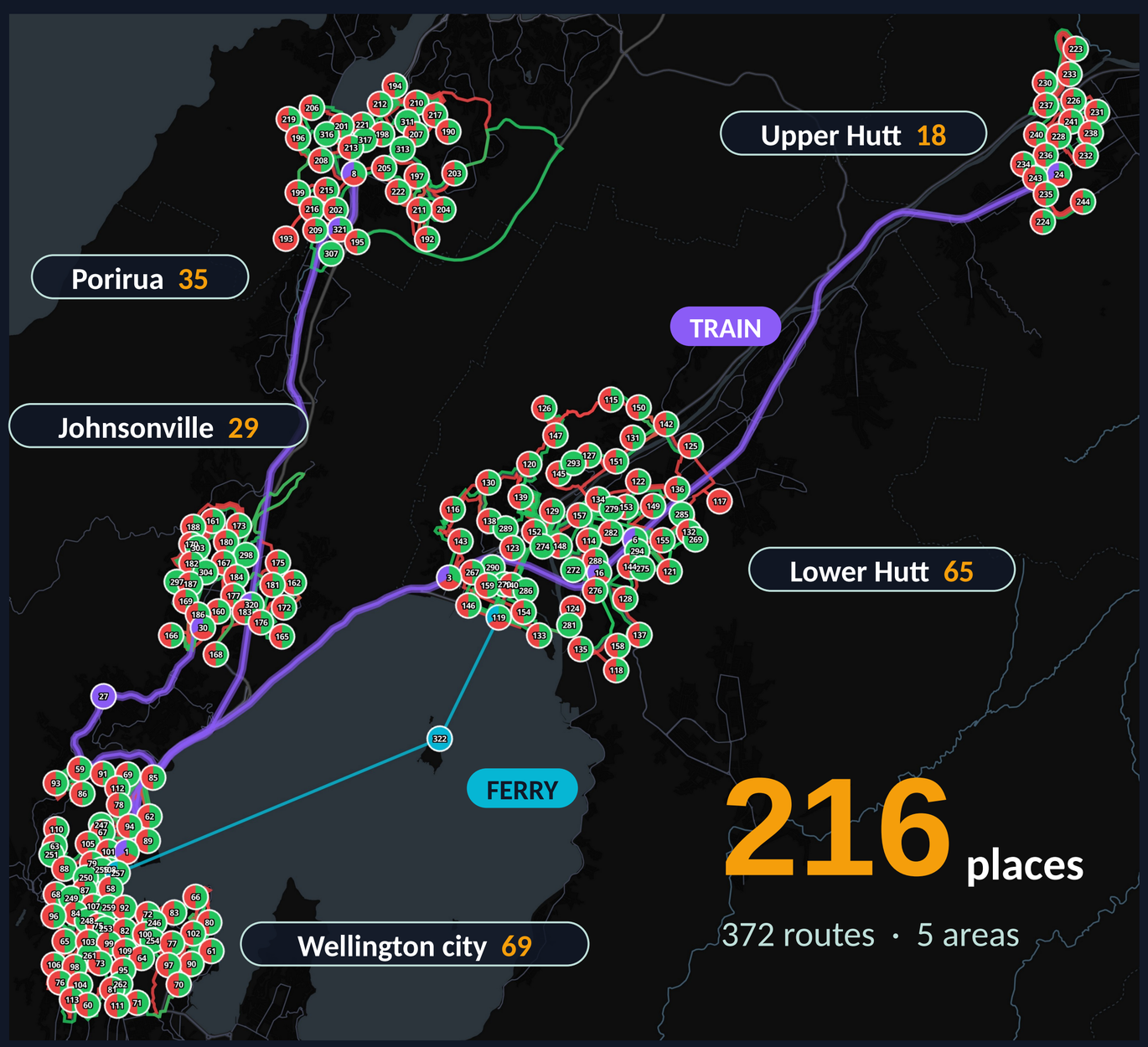
KEY FINDINGS

Open Day
played live by visitors
on the stand's laptops

Live play
every move appears on
every screen as it happens

No install
join from a web browser
with a link or a six-character code

THE BOARD: ALL OF WELLINGTON



WHY OPEN DAY?

- The old software demo was over five years old
- Visitors need something to try in minutes
- A map of their own city is instantly familiar
- It shows what a fourth-year student can build

HOW TO PLAY

- Mr X hides; up to five detectives hunt him
- Every move spends a travel ticket
- Mr X is shown only every few rounds
- Catch him within 24 rounds to win

Escooter · Bus · Train · Ferry · Invisible · Double

PLAY IT NOW
Scan or visit
vuw-mrx.xyz

TECHNICAL CHALLENGES

KEEPING A SECRET
Each player is sent their own view, so Mr X's position never reaches a detective's screen.

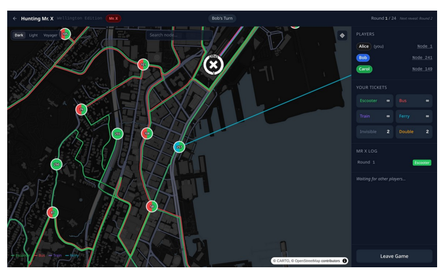
STAYING IN SYNC
Every move is pushed live to every screen. A dropped connection catches up by itself.

A FREE REAL MAP
Routes along real roads are worked out once, not paid for on every move.

PROVING IT WORKS
135 automated tests, including 300 random games. Every planted bug was caught.

TECH STACK

Screens	Vue 3 · TypeScript · Tailwind CSS
Map	MapLibre GL · OpenStreetMap
Server	Java 21 · Spring Boot
Live	WebSockets · STOMP
Testing	JUnit 5 · Mockito · PIT
Hosting	Docker · Cloudflare Tunnel


Mr X's screen in a live game

OUTCOME

- Played live at Open Day
- Anyone can join over the internet
- Ready for future Open Days

NEXT STEPS
- A big-screen view for the stand
- A phone-friendly layout
- Computer players for solo games

Board-game mechanics used with permission from Ravensburger, for non-commercial academic use.